Clinical care repeatedly draws on the same observations to answer different questions. A chest CT can inform several diagnoses, and physiological measurements can help interpret findings that imaging alone leaves uncertain. Developing a separate predictor for each clinical use often requires new labels and another round of training, even when the underlying evidence overlaps. Clinical foundation models offer a way to learn from that overlap: a shared model could acquire knowledge across patients and examinations, then adapt it to a new target with limited additional supervision. Clinical foundation models are especially valuable in medicine, where expert annotation is costly and the amount and type of clinical data vary from patient to patient. Realizing this promise requires understanding what information a model should preserve and how to make that information useful under changing clinical circumstances.

Consider a patient undergoing a routine chest CT. To assess this patient's cardiac health, clinicians may also need complementary information from ECG or echocardiography. A clinical foundation model could connect these sources by learning relationships across patients and examinations. Learning such relationships draws on records with different combinations of examinations and labels from automated interpretations, clinical reports, and expert review. These differences require the model to account for both the reliability of supervision and the availability of clinical evidence. Putting this knowledge into practice further requires adapting the model's predictions to the clinical task and the patient's preferences. Realizing the clinical value of multimodal learning therefore depends on reliable learning from imperfect evidence and adaptation to different clinical needs.

These challenges motivate my research on the learning and adaptation methods needed to build clinical foundation models. I pursue two complementary directions: (1) **learning from imperfect clinical data**; (2) **objective and preference adaptation**. The first addresses the quality and completeness of the evidence available for learning; the second addresses how models use that knowledge under different clinical goals and patient preferences. Across these directions, I develop theoretical foundations and computational methods for machine learning, and collaborate extensively with clinicians to formulate and investigate problems arising in medical practice.

SUMMARY OF PRIOR WORK

Imperfect supervision. Explicit models of label corruption can help recover information about the underlying target. I developed GNL [2], a theoretical framework that unifies generative and discriminative approaches to noisy-label learning through a direction-agnostic EM formulation. This framework provides a common foundation for modeling label corruption and learning the underlying prediction task. Across the evaluated benchmarks, it improved prediction and label-corruption estimation with less training compute than the compared baselines. My work also addresses complementary challenges in medical supervision. ACPL [3] selects informative unlabeled images and generates pseudo-labels to support learning with limited annotation. NVUM [4] uses long-term prediction memory to reduce the influence of noisy labels in imbalanced medical datasets. BoMD [12] uses multiple image descriptors aligned with label semantics to detect and refine noisy labels in multi-label chest X-ray classification.

Through collaboration with cardiologists, I identified a gap between clinical ECG interpretation and the supervision used to train diagnostic models. Automated systems provide initial interpretations, which cardiologists review and correct through expert over-reads. Models trained on the initial machine interpretations miss the expertise captured in these revisions. In our study [1], we compared machine and expert supervision using more than two million ECGs with paired interpretations. Expert-reviewed labels improved diagnostic performance across supervised learning, self-training, and multimodal contrastive learning, particularly for rare conditions.

Incomplete observations. Different input combinations reveal different aspects of a prediction target. A model must therefore share knowledge across these combinations while making use of the information available in each. Knockout [5] trains one predictor on different input subsets by replacing selected inputs with designated placeholders during training. The same placeholders mark missing inputs at deployment, allowing the model to predict from the available observations without imputing absent data or training a separate model for each input combination. When richer examinations are available during training, X-Cardia [8] transfers their information to routine chest CT by aligning CT representations with ECG- and echocardiography-derived measurements. To encourage models to learn from complementary inputs, ShapKO [7] adapts modality withholding according to each modality's estimated contribution.

In clinical screening, patients often have different examinations available. I study how models can use this incomplete information to identify additional disease risks through opportunistic screening. In Cardio Amyloid-AI [6], we combined routinely collected CT, ECG, echocardiographic, and demographic information to screen for transthyretin cardiac amyloidosis in patients with severe aortic stenosis. In the CT–LVEF study [9], we used non-contrast chest CT to identify abnormal LVEF, a functional measurement typically assessed with dedicated cardiac imaging. Together, these studies show how routinely collected information can support additional screening and help identify patients who may benefit from further cardiac assessment.

Objective and preference adaptation. Extending opportunistic screening to multiple conditions requires models that adapt shared imaging knowledge to different clinical objectives. In HyperCT [10], I developed a shared CT model with a hypernetwork that generates task-specific, low-rank parameter updates for pulmonary and cardiac prediction tasks. This allows the model to reuse imaging representations while specializing for each prediction target. HyperCT improved performance over the compared baselines while retaining parameter-efficient adaptation.

When several objectives must be optimized together, preferences specify how to balance them. My work on MAdam [11] examines whether the optimizer's updates preserve that intended balance. I showed that Adam's adaptive updates can distort the trade-off specified by a multi-objective solver, and developed a preference-conditioned curvature correction to account for this mismatch. This connects preferences over objectives with the updates used to train the model while preserving the underlying solver and optimizer.

FUTURE WORK

My goal is to build clinical foundation models that bring medical evidence together, adapt to new uses, and improve through collaboration with the people they serve. These are connected research problems: the model needs useful clinical knowledge, ways to apply it in different circumstances, and interaction that helps clinicians and patients guide its use and further development.

Multimodal Foundation Models for Healthcare. I will build models that connect different examinations into a shared representation of a patient's condition. An image can reveal anatomy, a physiological signal can describe function, and a clinical history can place both in the course of an illness. Learning their relationships could give a model a richer understanding than any one source provides. The scientific challenge is to capture what these observations explain together while preserving findings that only one modality reveals. I will develop learning objectives that connect complementary observations across partially overlapping records and account for differences in the quality of their supervision. The model should learn from patients with rich records and remain useful when a patient has only a subset of examinations, with uncertainty reflecting the evidence available. My work on imperfect supervision and

cross-modal learning provides the starting point. The broader aim is a medical representation that supports diagnosis, risk assessment, and monitoring across different combinations of clinical data.

Adaptive and Personalized Medical AI. I will develop ways to adapt a shared medical model to a new clinical use while retaining the capabilities it has already learned. A new diagnostic target may require different features; a new setting may provide different inputs; an individual decision may require a different balance of objectives. The challenge is to identify what should change in each case and preserve what remains useful. I will study task-conditioned components and learning procedures that selectively update a model, building on my work in conditional adaptation and multi-objective optimization. Personalization also requires a clear distinction between clinical evidence and individual priorities. Patient-specific observations should change risk estimates, while preferences should change how those estimates inform choices among clinically appropriate options. I will investigate how a shared outcome model can support these different choices and how uncertainty should affect a recommendation. This direction will make specialization and personalization capabilities of the medical model, allowing its knowledge to serve new tasks and individual needs.

Human–AI Collaboration in Healthcare. I will develop medical AI that helps clinicians and patients examine a decision together and learns from the expertise and priorities they contribute. The model can organize relevant evidence and compare possible interpretations; clinicians can correct its understanding, and patients can explain which consequences matter to them. The scientific challenge is to understand what this interaction teaches the model. Correcting a finding changes the evidence for a prediction, while clarifying a preference changes how that prediction should inform a choice. I will study ways to elicit and learn from these distinct forms of feedback, including when the model should ask a question or bring uncertainty to a person's attention. Evaluation will examine whether the interaction improves joint decisions and whether the feedback supports model improvements beyond the individual case. My work with cardiologists on expert ECG supervision establishes a basis for studying the contribution of clinical expertise. This program extends that work toward collaboration in which people actively guide both the model's immediate assistance and its subsequent learning. Feedback from these interactions will guide improvements in both the model's medical knowledge and its adaptation to new uses.

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